

About Me

AI/ML enthusiast with hands-on experience building and deploying full-stack, data-driven applications. Proven ability to improve model performance, automate complex tasks, and deliver usable solutions. Strong foundation in NLP, computer vision, and modern web technologies. Interested in building reliable AI systems that connect models, APIs, and user-facing applications.

Education

B.S. Computer Science

2021-2025

FAST National University of Computer and Emerging Sciences (FAST-NUCES), Islamabad

Experience

AI Engineer

Dec 2025 – Present

Oak Street Technologies

Islamabad, Pakistan

- Support AI and automation work around Enwage, a cloud-based HCM and payroll platform, focusing on ways to make HR and workforce processes simpler and more data-driven.
- Helped build chatbot features for Enwage's staffing and staging platforms, improving how users retrieve the information they need while adding guardrails for safer and more controlled responses.
- Worked on enhancing a VectorDB + RAG pipeline by improving retrieval quality, context relevance, and response grounding to support more accurate and reliable downstream outputs; used LangSmith for tracing and debugging workflow behavior.

Software Engineer

Aug 2025 – Dec 2025

EUNOMATIX

Austin, Texas, USA

- Contributed to full-stack solutions for web portals and backend services on projects such as **BrowserMon** and **Watchdog** for browser-activity monitoring.
- Worked on improving how BrowserMon and related monitoring tools identify potentially harmful or suspicious websites by refining detection logic, signals, and supporting workflows.
- Built small automation scripts and internal tools in Python to support data flows between services, simplify operational tasks, and improve repeatable workflows.
- Worked with an Agile team to ship incremental improvements, participate in code reviews and keep basic technical documentation up to date.

Cloud Application Development Intern

June 2023 – Aug 2023

Systems Limited

Islamabad, Pakistan

- Designed and deployed 3 cloud-based dashboards using React.js and MS SQL Server, enabling 100+ users to visualize sales and HR analytics.
- Reduced API response time by 30% by implementing optimized queries and client-side caching.
- Worked with a 5-member Agile team to build dashboards and improve data accuracy, performance, and UI consistency.

Projects

InfoMorph – Multilingual News Summarizer

- Developed an NLP-based web app that summarizes 50+ real-time news articles in Urdu and English using the Pegasus Transformer model.
- Achieved 65% average summary compression while maintaining over 85% content integrity through preprocessing, model tuning, and output evaluation.
- Implemented a scalable backend using FastAPI and MongoDB with async scraping, API-driven processing, and support for 50+ concurrent users in testing.
- Frontend built with Next.js; integrated Eleven Labs for multilingual speech synthesis with First Order Motion Model used for AI News Anchor.

CNN Image Classifier

- Trained a 6-layer CNN using Keras on 25,000 labeled images from Kaggle's Dogs vs Cats dataset; reached 96% accuracy on validation set.
- Improved generalization by applying data augmentation (rotation, flipping) and dropout (rate = 0.5), reducing overfitting by 18%.
- Exported the trained model for deployment in other environments, improving inference speed by 20% on local devices.

Urdu-to-English Neural Translator

- Created a neural machine translation system using PyTorch with an LSTM-based Seq2Seq model and Bahdanau Attention; trained on 50,000+ sentence pairs.
- Boosted BLEU-4 score by 14 points through beam search (width = 3), token masking, and teacher forcing.
- Optimized the preprocessing pipeline with padding and custom tokenization, cutting training time by 22%.
- Deployed as a Streamlit app with language switcher and live inference for 20+ test users.

Skills

Programming: Python, JavaScript, SQL, C++, Java**AI / ML:** PyTorch, TensorFlow, Scikit-learn, Transformers, LSTM, Seq2Seq, BERT, BLEU, RAG**Tools & Web:** FastAPI, LangChain, LangGraph, LangSmith, MongoDB, MySQL, Git, Docker, Jupyter, Linux, React.js, Next.js, Node.js, REST APIs